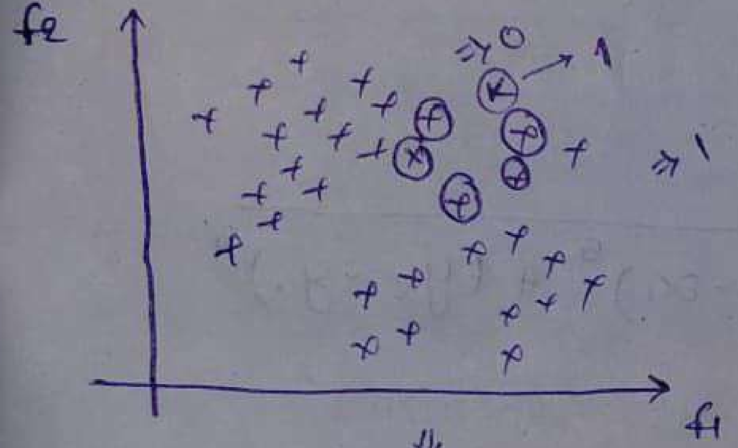
K Nearest Neighbour (KNN)

1. Classification
2. Regression

1. Classification



⇓

TRAINING

[Binary category]

f1	f2	y
–	–	0
–	–	1

0's → 2

1's → 3

$\boxed{K = 5}$

1. We have to initialize the K value

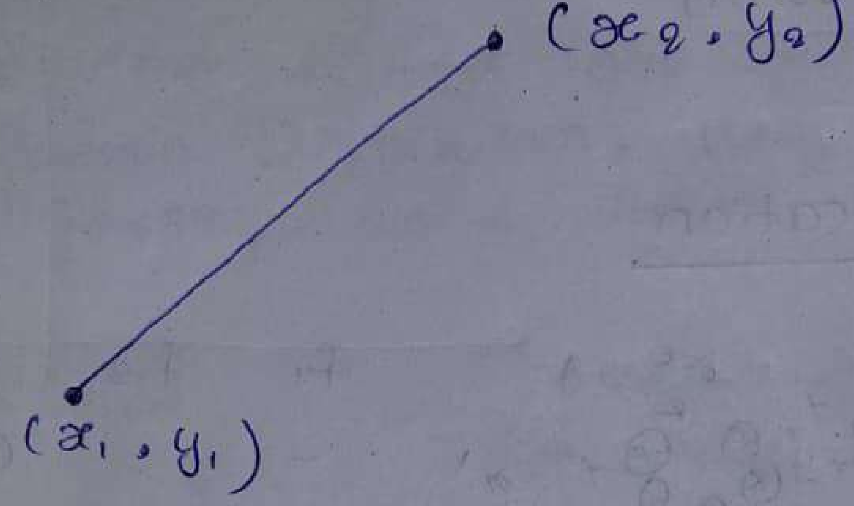
   $K > 0 \ldots\ldots\ldots \infty$

   $K = 1, 2, 3, 4, 5, \ldots\ldots\ldots$ ⇒ Hyperparameter

2. Find the K nearest Neighbour for the test Data.

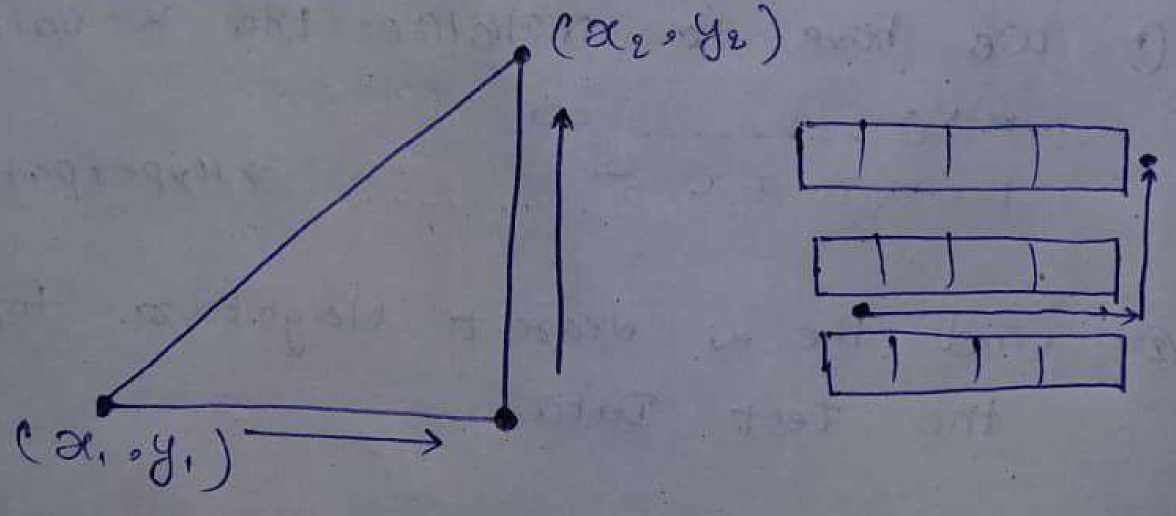
3. From those $K = 5$ how many neighbour belong to 0 category and 1 category

(a) Euclidian Distance

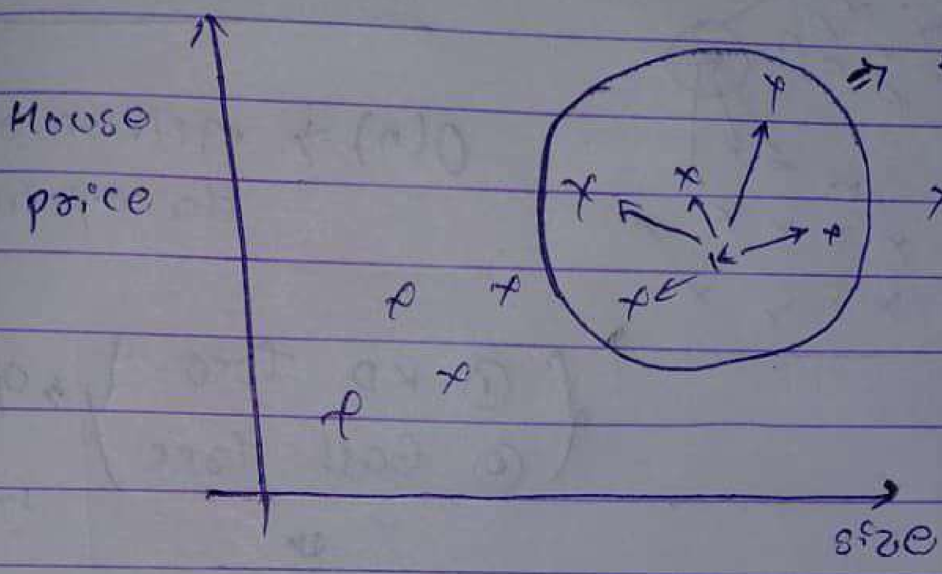


$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(b) Manhattan Distance



(2) Regression

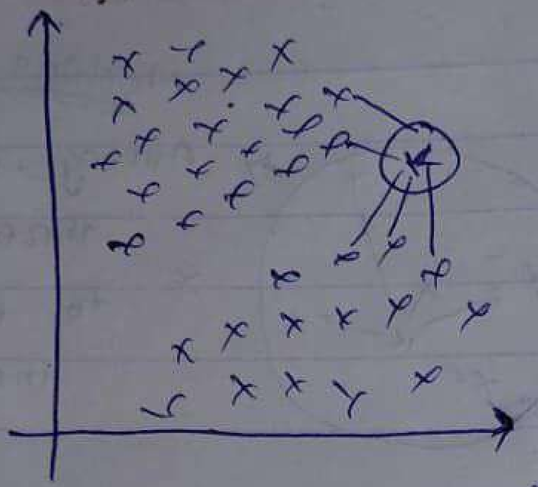


Median

⇒ Average of all these point's to find out the O/P

Note: All the steps are same as well as classification find K value ~~then~~ based on the neighbour.

* find out the average or you have lots of outlier you can go also with the median.



Time complexity

$O(n) \Rightarrow$ million of datapoint

{ ① KD tree, ② Ball tree } $\Rightarrow$ optimize

$\Downarrow$

Binary Tree

Note: In Binary Tree it to calculate the distance between in every point in the test data is reduce by half because it will not be checking with every respect data point.

That is the power of Binary Tree.